

SUSTAINABLE COMMUNITY DEVELOPMENT THROUGH PEER-TO-PEER LEARNING

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Abstract

When starting a new skill, beginners often struggle a lot because they don't have experienced people to guide them. Today, most online learning is just watching videos by yourself, which lacks the real human interaction people actually need to improve. To fix this gap, our app connects beginners who want to learn with veterans who are willing to share their knowledge. The app covers a wide variety of skills, and users can learn and build communities at any time of the day. Because it is online, you can use it anywhere depending on what skill you are practicing. It is made for everyone. If you are a beginner, you can just browse the app for a skill you like and join a group. On the flip side, if you are an expert who wants to help others, you can easily create your own group on the app and start guiding people right away.

Keywords:

Community development, peer-to-peer learning, learning by doing.

I. INTRODUCTION:

Digital learning made information universally accessible, but accessibility doesn't always guarantee mastery. This is generally experienced by Self-taught learners, hobbyists, and professionals looking to upskill; consuming one-way content (like pre-recorded videos or text modules, YouTube tutorials, etc.). The gap is felt mostly when a learner makes a mistake or encounters a bug they cannot figure out how to fix on their own or when initial excitement fades, and the isolation of learning alone causes motivation to drop or when the learner tries to apply the knowledge acquired in real-life scenarios and needs a feedback on his/her work.

To address this, we propose a community application where people can come forward to help others learn skills – for free. Instead of formal, high-pressure teaching, it relies on shared routines.

II. LITERATURE REVIEW:

A recent study states there's a critical role of interactive, peer-to-peer learning activities in overcoming limitations faced by traditional, isolated e-learning. Campbell explains how peer-to-peer learning helps in community development as the learners actively support one another.

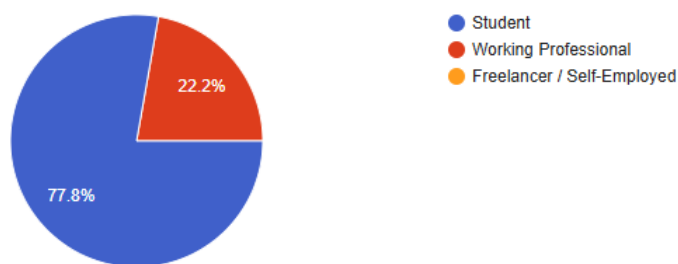
The necessity of this community-driven approach becomes even clearer when examining the psychological drawbacks of unguided online learning. Digital learning without human interaction leads to increased feelings of loneliness, low attention span, etc. Effective modern learning must go beyond this isolated content consumption. Our application fixes these problems with a digital interface strictly as a discovery tool to initiate physical, peer-to-peer skill-sharing sessions, transforming passive learners into active, supported community participants.

III. SURVEY ANALYSIS:

To understand the problems faced by learners and to evaluate the proposed peer-to-peer learning application, I conducted a public survey via google forms. A total of 45 responses were collected. The survey examined respondents' current status, their use of online learning resources, difficulties experienced while learning independently, willingness to participate in the proposed "tag along" model, and their overall opinion of the application. The analysis below presents the quantitative results along with qualitative observations from the open-ended responses.

4.1 CURRENT STATUS OF RESPONDENTS:

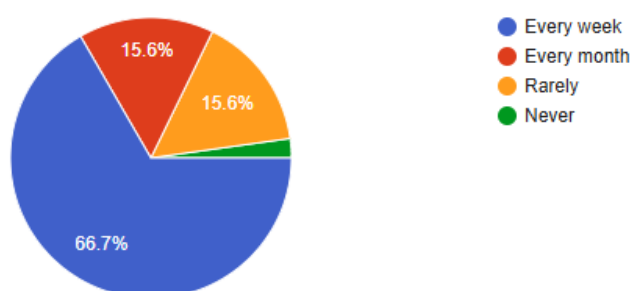
The first question was used to understand the general background of the respondents. As the graph presents, majority of the respondents were students.



Out of the 45 respondents, 35 respondents (77.8%) were students, while 10 respondents (22.2%) were working professionals. No respondent selected the Freelancer / Self-Employed or Other categories.

4.2 FREQUENCY OF INTERNET-BASED LEARNING:

The second question examined how frequently respondents use resources such as YouTube, online courses, and blogs to learn a new skill.



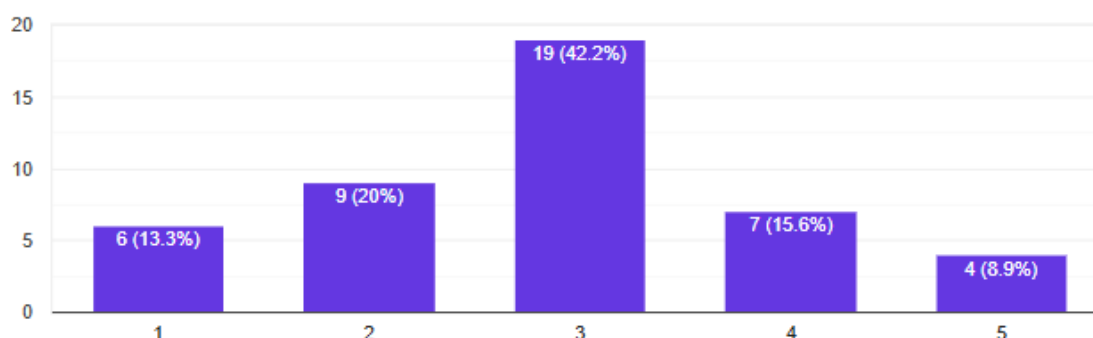
Among the 45 respondents, 30 respondents (66.7%) use the internet every week to learn a new skill. 7 respondents (15.6%) use online resources every month, another 7 respondents (15.6%) use them rarely, and 1 respondent (2.2%) reported never using them.

Therefore, 37 respondents (82.2%) use internet-based resources at least monthly, while two-thirds of the respondents use them every week. This demonstrates that digital learning is already widely accessible and commonly used within the surveyed population.

4.3 LOSS OF MOTIVATION DURING INDEPENDENT LEARNING:

The third question asked respondents to rate from 1 to 5 how often they lose motivation or feel like giving up when trying to learn something completely on their own. The responses were concentrated around the middle of the scale.

45 responses

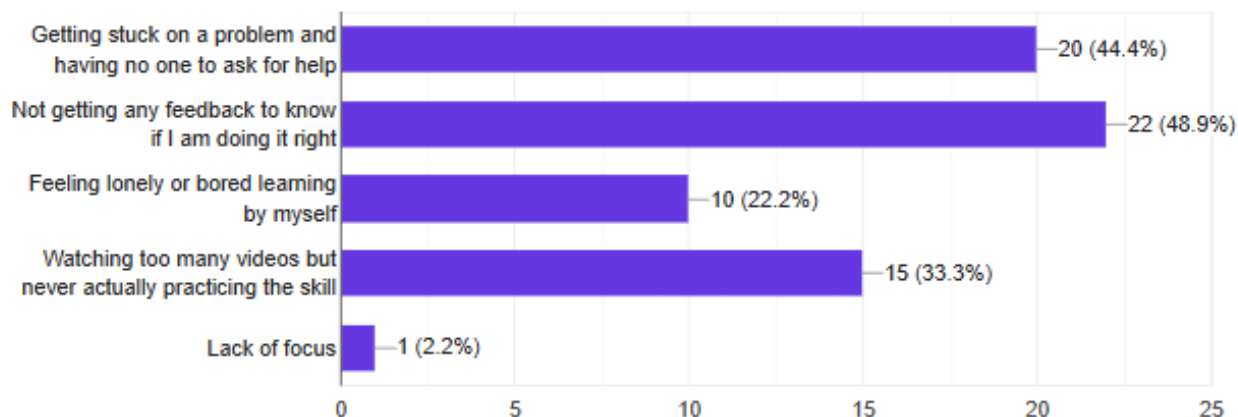


approximately two-thirds of the sample reported at least a moderate tendency to lose motivation when learning completely independently. Furthermore, 11 respondents (24.4%) selected either 4 or 5, representing relatively frequent or very frequent loss of motivation.

These results provide quantitative support for the motivational aspect of the Guidance Gap described in this research. Although independent learning does not negatively affect every respondent to the same extent, the majority experiencing at least a moderate level of motivational difficulty indicates that access to information alone may not be enough to sustain the learning process. A peer-to-peer environment could potentially address this issue by providing learners with human interaction, encouragement, accountability, and someone to learn alongside.

4.4 PROBLEMS FACED WHILE LEARNING ONLINE:

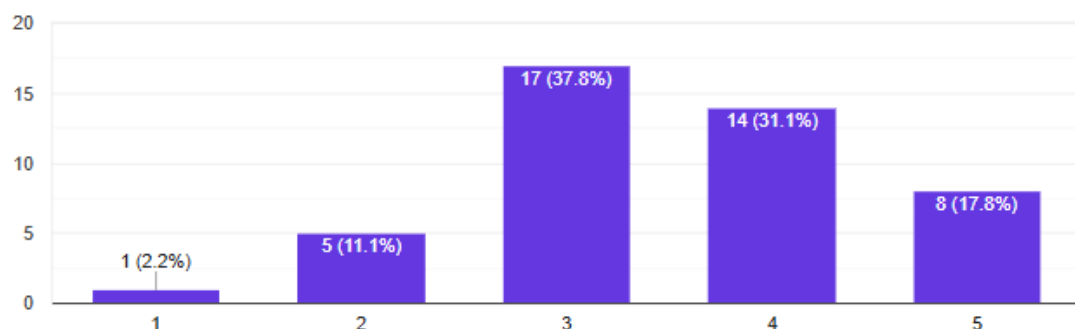
The fourth question examined the main problems respondents experience while learning online. Respondents could select more than one option; therefore, the percentages do not add up to 100%. The most common problems were related to feedback and access to help.



The most frequently reported problem was not receiving feedback, selected by 22 respondents (48.9%). This was followed closely by getting stuck on a problem without anyone to ask for help, reported by 20 respondents (44.4%). 15 respondents (33.3%) reported watching too many videos without actually practicing the skill, while 10 respondents (22.2%) reported feeling lonely or bored while learning alone. One respondent additionally identified lack of focus as a problem.

4.5 LIKELIHOOD OF USING THE "TAG ALONG" MODEL:

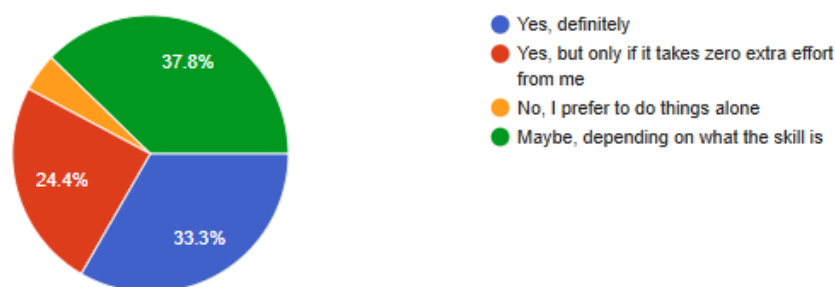
The fifth question introduced the main concept of the proposed application. Respondents were asked how likely they would be to use a system where beginners can find experienced individuals, or "veterans", and tag along while they perform their normal routines. The overall response was favourable.



The average likelihood score was 3.51 out of 5. A total of 39 respondents (86.7%) selected a rating of 3 or above, while 22 respondents (48.9%) selected either 4 or 5. Only 6 respondents (13.3%) selected 1 or 2. This result provides preliminary evidence of interest in the proposed peer-to-peer model. Almost half of the respondents expressed a high likelihood of using the application, while a large majority showed at least moderate interest. This supports the idea that learners may be willing to move beyond passive consumption of tutorials toward an approach based on observing and interacting with experienced people in their normal routines.

4.6 WILLINGNESS TO ACT AS A VETERAN:

The sixth question examined the other side of the proposed community. Respondents were asked whether, if they were experts at something, they would allow a beginner to tag along as long as the interaction did not require additional time from them. The majority were open to the idea.

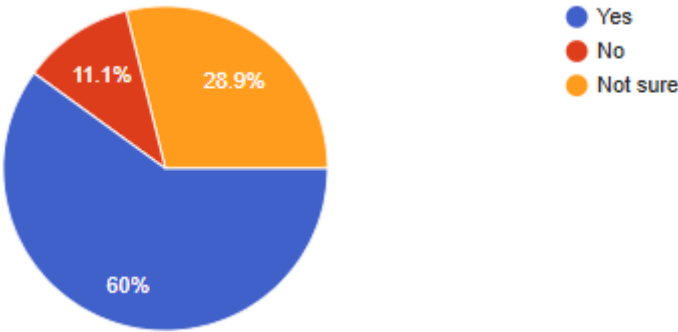


A total of 26 respondents (57.8%) gave an explicit positive response, either by selecting "Yes, definitely" or agreeing provided that the interaction required no additional effort. Another 17 respondents (37.8%) were potentially willing, depending on the skill. Only 2 respondents (4.4%) stated that they preferred to do things alone.

This finding is important because the proposed application requires participation from both sides of the learning relationship. The results indicate that there is potential not only for demand from beginners but also for a supply of experienced individuals willing to share their existing routines. The fact that many respondents emphasized zero additional effort also supports the proposed design principle of shared routines, where veterans do not have to prepare formal lessons but can allow beginners to observe their existing activities.

4.7 REAL-LIFE OBSERVATION VERSUS VIDEO LEARNING:

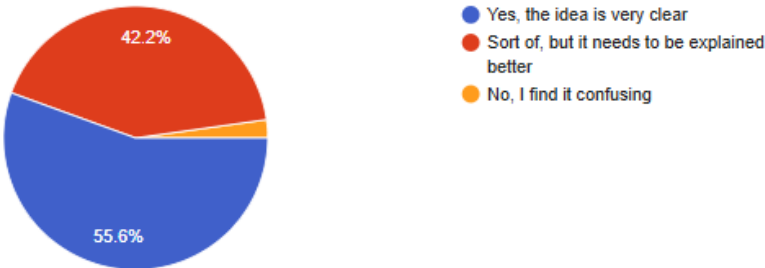
The seventh question tested one of the central assumptions behind the proposed application: whether observing someone perform a skill in real life could help a learner more than simply watching a video. The majority of respondents believed that real-life observation could be beneficial.



27 respondents (60.0%) answered Yes, indicating that they believed watching someone perform a skill in real life could help them learn faster than watching a video. 13 respondents (28.9%) were not sure, while only 5 respondents (11.1%) answered No. The result provides support for the experiential component of the proposed model. A majority of respondents perceive value in seeing a skill performed in a real-life context rather than relying entirely on video-based instruction. This connects directly with the paper's concept of learning by doing, where the learner can observe how knowledge is applied in an actual situation and potentially ask questions or receive immediate feedback. However, this result measures respondents' perceptions rather than objectively measured learning speed. Therefore, it supports the perceived usefulness of real-life observation but does not by itself establish that real-life observation produces better learning outcomes than video learning.

4.8 OVERALL UNDERSTANDING OF THE COMMUNITY-LEARNING CONCEPT:

The eighth question asked respondents whether the idea of an application that builds a community by helping people learn from each other made sense to them. Almost all respondents were able to understand the basic concept.



25 respondents (55.6%) stated that the idea was very clear. A further 19 respondents (42.2%) understood the general concept but believed that it needed to be explained better. Only 1 respondent (2.2%) found the concept confusing. Therefore, 44 respondents (97.8%) either clearly understood the idea or understood it sufficiently to recognize the concept. This provides strong preliminary evidence that the basic peer-to-peer community model is understandable to potential users.

At the same time, the relatively large proportion of respondents who selected "Sort of" highlights an important area for improvement. The application needs to clearly communicate how the interaction actually takes place, particularly the difference between traditional online classes and the proposed model of discovering and joining real-world or live peer-to-peer skill-sharing sessions. This is particularly relevant because the paper proposes that the digital interface should primarily function as a discovery tool for initiating peer-to-peer sessions.

4.9 ADDITIONAL THOUGHTS, SUGGESTIONS AND CONCERNS:

The final question was open-ended and allowed respondents to provide additional thoughts, suggestions, or concerns about the proposed application. Of the 45 respondents, 18 respondents (40.0%) left the question blank. Another 13 respondents (28.9%) provided short non-substantive responses, such as "No", "NA", "No suggestions", "idk", or "Nope thanks". The remaining 14 respondents (31.1%) provided substantive comments, suggestions, or concerns.

The responses highlighted several important themes, including trust, safety, verification, and clarity of the platform's functionality. Respondents suggested features such as verified profiles, ratings and reviews, scheduling, AI-based skill matching, study groups, and rewards for mentors. Concerns were also raised about fake or inactive users, interacting with strangers, user safety, and whether the platform would involve live, one-to-one, or pre-recorded learning. Some respondents questioned its suitability across different age groups and the overall ease of participation. These findings indicate that while the peer-to-peer concept is positively perceived, its success will depend on creating a safe, trustworthy, clearly structured, and accessible community. Thus, addressing the Guidance Gap requires not only connecting learners with experienced individuals but also providing an environment that encourages meaningful and reliable peer-to-peer interaction.

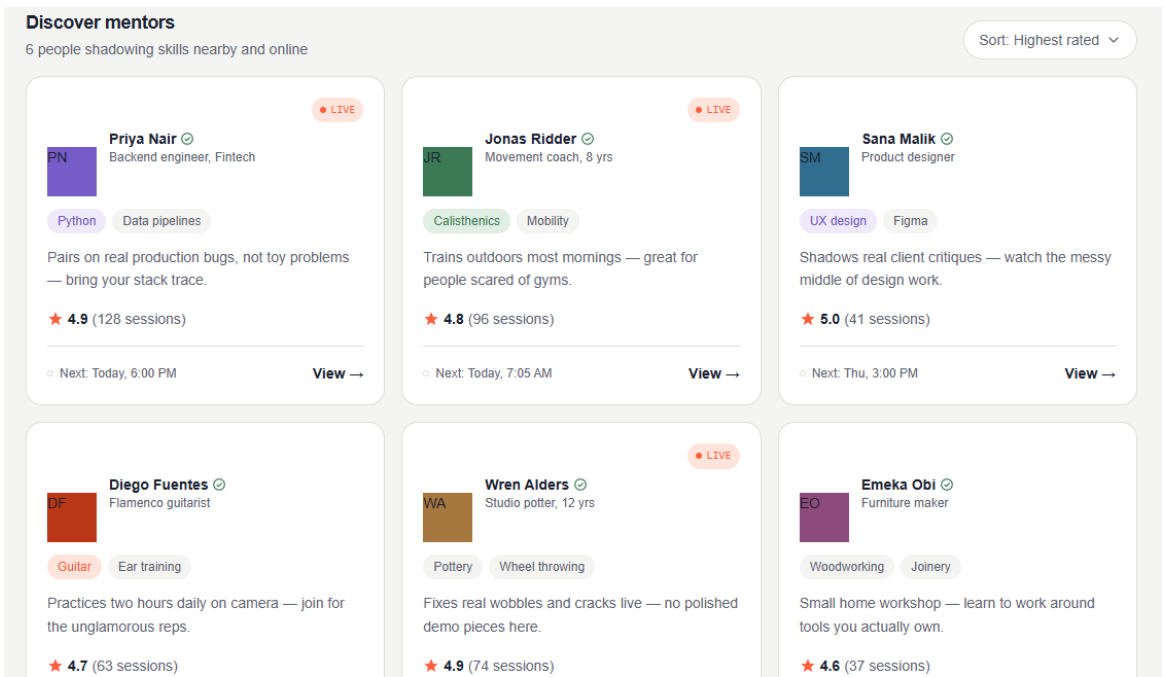
IV. IMPLEMENTATION:

The proposed system is a community-driven web application that connects beginners (learners) with experienced individuals (veterans) for real-world and live-digital skill shadowing. The platform primarily acts as a digital hub for discovering suitable people, scheduling sessions, communicating with participants, and maintaining user reputation. Unlike conventional learning platforms, it does not depend on pre-recorded instructional content or require mentors to prepare formal lessons.

The process begins with user registration and verification. Veterans can provide relevant evidence of their skills, such as previous work or professional profiles, to establish credibility. Learners can then search for veterans based on skill, proficiency level, location, or time zone. Veterans list activities according to their existing routines, and learners can request to join suitable sessions. A communication feature can be used to coordinate the session before it takes place.

During the shadowing session, the learner observes and interacts with the veteran while the skill is being performed naturally. Depending on the skill, the session may take place physically, such as during a fitness or photography activity, or digitally through live video or screen sharing for activities such as programming and data analysis. The learner can ask questions and participate where appropriate, creating a more active learning experience than simply watching pre-recorded content.

After each session, a two-way rating and review system allows both participants to provide feedback. Learners can evaluate veterans based on helpfulness and the quality of the interaction, while veterans can provide feedback on learner behaviour and participation. Verification, reputation, reporting, and safety mechanisms can help maintain trust within the community. Overall, the implementation focuses on minimizing additional effort for veterans while providing beginners with practical exposure, interaction, and guidance.



V. CONCLUSION:

This research identifies the Guidance Gap in conventional online learning, where learners have access to abundant information but may lack feedback, practical guidance, motivation, and human interaction. The survey findings indicate that these difficulties

are experienced by a significant proportion of respondents, while the proposed peer-to-peer "tag along" model received generally positive responses.

The proposed application addresses this gap by connecting beginners with experienced individuals through their existing routines rather than requiring formal teaching or pre-recorded content. By combining skill discovery, scheduling, live shadowing, and community-based feedback, the platform aims to transform passive online learning into a more active, practical, and socially connected learning experience. Although challenges such as safety, verification, trust, and user participation remain, the proposed model provides a potential approach for sustainable peer-to-peer skill development and community learning.

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